Super Key and Candidate Key

Finding Super Keys and Candidate Keys

Relation

STUDENT(StudentID, RollNo, Name, Department)

StudentID	RollNo	Name	Department
S101	11	Amit	CSE
S102	12	Priya	CSE
S103	13	Rahul	ECE
S104	14	Neha	ME

Find Super Keys

Since **StudentID** uniquely identifies every row:

Super Keys:

- {StudentID}
- {StudentID, Name}
- {StudentID, RollNo}
- {StudentID, Department}
- {StudentID, Name, RollNo}
- {StudentID, Name, Department}
- {StudentID, RollNo, Department}
- {StudentID, RollNo, Name, Department}

Similarly, since **RollNo** is also unique:

- {RollNo}
- {RollNo, Name}
- {RollNo, Department}
- {RollNo, StudentID}
- {RollNo, StudentID, Name}
- {RollNo, StudentID, Department}
- {RollNo, Name, Department}
- {RollNo, StudentID, Name, Department}

Candidate Keys

{StudentID}, {RollNo}

Relation

EMPLOYEE(EmpID, AadhaarNo, Name, Salary)

EmpID	AadhaarNo	Name	Salary
E101	123456789111	Raj	45000
E102	123456789112	Anu	50000
E103	123456789113	Raj	47000
E104	123456789114	Ravi	52000

Super Keys

Using **EmpID**

- {EmpID}
- {EmpID, Name}
- {EmpID, Salary}
- {EmpID, AadhaarNo}
- {EmpID, Name, Salary}
- {EmpID, AadhaarNo, Name}
- {EmpID, AadhaarNo, Salary}
- {EmpID, AadhaarNo, Name, Salary}

Using **AadhaarNo**

- {AadhaarNo}
- {AadhaarNo, Name}
- {AadhaarNo, Salary}
- {AadhaarNo, EmpID}
- etc.

Candidate Keys

{EmpID}, {AadhaarNo}

PATIENT(PatientID, PhoneNo, Name, City)

PatientID	PhoneNo	Name	City
P101	9876543210	Ravi	Delhi
P102	9876543211	Priya	Mumbai
P103	9876543212	Ravi	Delhi
P104	9876543213	Mohan	Chennai

Super Keys

Using PatientID
- {PatientID}
- {PatientID, Name}
- {PatientID, City}
- {PatientID, PhoneNo}
- {PatientID, Name, City}
- etc.

Using PhoneNo
- {PhoneNo}
- {PhoneNo, Name}
- {PhoneNo, City}
- etc.

Candidate Keys

{PatientID}, {PhoneNo}

BOOK(BookID, ISBN, Title, Author)

BookID	ISBN	Title	Author
B101	978001	DBMS	Navathe
B102	978002	AI	Russell
B103	978003	Python	Mark
B104	978004	DBMS	Navathe

Super Keys

Using BookID

- {BookID}
- {BookID, Title}
- {BookID, Author}
- {BookID, ISBN}
- etc.

Using ISBN

- {ISBN}
- {ISBN, Title}
- {ISBN, Author}
- {ISBN, BookID}
- etc.

Candidate Keys

{BookID}, {ISBN}